

M16 Review / Repair - Teaching RC2

OPEN ONLY AFTER SAVED ATTEMPT

P01 repair - Production mindset

A data engineer owns behavior over time under failures, changes and other users, not only a correct local output.

Repair principle: Production-ready is a spectrum with explicit risks/SLA, not a magic label.

Evidence focus: recoverable -> rollback/rebuild/runbook

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P02 repair - Testing strategy for data engineering

Different bugs require different tests; one giant end-to-end test is slow and gives poor diagnosis.

Repair principle: Unit/pure transformation tests use tiny controlled inputs.

Evidence focus: smoke -> deployment/system wiring

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P03 repair - Automated quality gates

A rule that logs ERROR but still publishes bad data is not a gate.

Repair principle: Define blocking invariants vs warnings with owner/SLA.

Evidence focus: optional field fill-rate -> monitor/warn

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P04 repair - Continuous Integration fundamentals

CI reduces the chance one developer's change silently breaks shared pipelines.

Repair principle: CI trigger often pull request/push; exact policy depends repo.

Evidence focus: secrets -> scoped test secret if truly needed

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P05 repair - GitHub Actions for data projects

Workflow YAML is executable automation tied to repository events; unsafe permissions/actions/secrets can affect supply-chain security.

Repair principle: Workflow has triggers, jobs, runners and steps.

Evidence focus: failure -> job red/non-zero

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P06 repair - Dev, test and prod environments

Environment separation limits blast radius and makes testing realistic without letting experiments modify production.

Repair principle: Code artifact/version should be promoted; environment config/connection endpoints differ.

Evidence focus: prod -> live controlled operations

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P07 repair - Deployment and rollback

Code rollback can be simple; data migrations/writes may be irreversible without an explicit recovery plan.

Repair principle: Deploy immutable/versioned artifact/commit, not "whatever is on laptop".

Evidence focus: credential change -> rotation/connection update

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P08 repair - Secrets and credential safety

Credentials grant access; once exposed in Git/log/screenshot, you must assume someone could copy them even if you delete the latest file.

Repair principle: Secrets include passwords, API tokens, private keys, SAS URLs/credentials.

Evidence focus: verify -> old credential no longer works

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P09 repair - RBAC and least privilege

Over-broad access increases blast radius of bugs and credential compromise.

Repair principle: RBAC maps roles/identities to permitted actions/resources.

Evidence focus: prod operator -> deploy/incident scopes, not all data by default

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.

P10 repair - PII, encryption and masking

Security starts by not storing/accessing sensitive data unnecessarily. Encryption alone does not make unrestricted plaintext use safe.

Repair principle: PII/sensitive classes depend on jurisdiction/policy; identify direct/quasi identifiers and business sensitivity.

Evidence focus: analytics aggregate -> drop/minimize identifier

STOP - check your understanding

Close Review and answer a changed version.

Answer in your own words before continuing.